

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY
Faculty of Information Technology I · Computer Science Department

NITS Research Residency Program

Nurturing IT Student Talent

Partner Introduction Document

2026 Inaugural Cohort
Hanoi, Vietnam

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1. Executive Summary

The **NITS Research Residency Program** is a selective, long-term talent development initiative hosted by NITS Lab — *Nurturing IT Student Talent* — within the Faculty of Information Technology I at the Posts and Telecommunications Institute of Technology (PTIT), Hanoi, Vietnam. The program identifies the most exceptional undergraduate computer scientists from a pool of more than one thousand Faculty of IT students and pairs them with leading researchers and industry professionals worldwide for a structured apprenticeship of up to two years.

The program operates across two tracks: a **Research Track** designed to produce serious scholars — culminating in a top-tier international publication — and an **Engineer Track** aimed at equipping residents to excel at the world's leading technology companies. In both tracks, residents are carefully selected through a five-stage vetting process and remain accountable to clear milestones throughout their tenure.

We invite universities, research institutes, and technology companies to engage as **Research Mentors**, **Industry Partners**, or **Program Sponsors**, contributing to the next generation of Vietnamese computer science talent while building a sustainable pipeline of well-prepared collaborators.

2-Year RESEARCH TRACK	1,000+ CANDIDATE POOL	5-Stage SELECTION PROCESS	Tier-1 PUBLICATION TARGET
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2. About PTIT and NITS Lab

2.1 Posts and Telecommunications Institute of Technology

PTIT is one of Vietnam's leading technical universities, established in 1953 and operating under the Ministry of Information and Communications. With campuses in Hanoi and Ho Chi Minh City, PTIT trains thousands of engineers and computer scientists each year. Its Faculty of Information Technology I houses several hundred of the most competitive IT undergraduate and graduate students in Vietnam.

Founded	1953
Ministry	Ministry of Information and Communications, Vietnam
Location	Hadong Campus, Hanoi, Vietnam

Faculty of IT I	One of Vietnam's premier IT faculties, 1,000+ undergraduate students
Graduate Programs	Active M.Sc. and Ph.D. programs in Computer Science

2.2 NITS Lab — Nurturing IT Student Talent

NITS Lab is the flagship talent development initiative of the Computer Science Department within the Faculty of IT I. Founded and directed by Dr. Long Hoang Dang, the Lab operates as a structured ecosystem rather than a traditional research group: it offers students a clear developmental pathway from foundational skills through to publication-quality research and industry readiness.

The Lab's philosophy is that *exceptional outcomes require exceptional environments*. Students are not simply assigned projects — they are mentored, held to high standards, connected to the global research community, and given the scaffolding to become serious contributors to their fields within their undergraduate years.

Key programmes and activities of NITS Lab include:

- **NITS Research Residency:** The selective two-track research and engineering programme described in this document.
- **Advanced Algorithms Programme:** Intensive technical training for years 2–3, including regular competitive programming contests.
- **AI Certification Pathway:** Structured certifications from NVIDIA, Microsoft, AWS, and other leading platforms.
- **International Summer Schools:** Co-organized with global partners including Singapore Institute of Technology and Brunel University London.
- **Industry Connections:** Partnerships with FPT, Viettel, Microsoft Vietnam, Samsung, and other leading companies for internships, projects, and graduate placement.

3. The Residency Programme

3.1 Overview

The NITS Research Residency is modelled on the structures of leading industry research residencies — such as those at VinAI and FPT AI — adapted for the academic setting of a top Vietnamese university. It provides a small cohort of selected undergraduates with sustained, high-quality mentorship and a research environment that demands and rewards genuine scholarly output.

Residents commit to the programme for its full duration and are subject to periodic review. Those who do not meet the programme's standards — in terms of commitment, progress, or quality — may be asked to leave. This rigorous approach is what distinguishes the Residency from conventional undergraduate research opportunities.

The NITS Research Residency is not a short-term training programme. It is an environment for students who want to develop seriously — as researchers or as engineers — and are willing to commit the time and effort that path requires.

3.2 Two Tracks

Research Track (approx. 2 years)

The Research Track is designed for students with strong mathematical foundations, research curiosity, and long-term aspirations in academia or applied research. Residents in this track work directly with a faculty or external mentor on a substantive research question, with the expectation of producing a publication at a competitive international venue by the end of their tenure.

Expected Output: A paper submitted to or accepted by a venue indexed in ISI / Scopus, co-developed with the resident's mentor.

Engineer Track (approx. 1 year)

The Engineer Track is designed for students targeting positions at major technology companies — both Vietnamese (VNG, FPT, Viettel) and international (Google, Microsoft, Amazon, Samsung, and others). Residents in this track participate in real industry projects with the Lab's partner companies and are trained intensively in system design, coding interviews, and software engineering best practices.

Expected Output: A completed industry project with a partner company, a strong portfolio, and documented progress toward a full-time or internship offer from a technology company.

3.3 What Makes This Programme Distinctive

FOR RESEARCH PARTNERS

- › Access to rigorously pre-screened, highly motivated undergraduates
- › Lab faculty provide day-to-day supervision — partners focus on research direction
- › Structured milestone framework ensures accountability and output quality
- › A natural pipeline of well-prepared future graduate applicants
- › Opportunity to build an ongoing institutional partnership with PTIT

FOR INDUSTRY PARTNERS

- › Early access to Vietnam's most technically competitive undergraduates
- › Residents trained to the standard of leading global tech companies
- › Flexible engagement models: projects, internships, co-supervision
- › Branding and visibility to a high-potential talent pool
- › Pathway to formal MOU and longer-term talent collaboration with PTIT

4. Resident Profile and Selection

4.1 Who Are Our Residents

Residents are recruited from the second, third, and — where appropriate — fourth year of PTIT's Faculty of Information Technology I undergraduate programmes. The inaugural cohort draws from a pool of more than **one thousand students** across the Faculty — already among the most competitive IT undergraduates in Vietnam.

We look for students who demonstrate:

- **Academic excellence:** Consistently strong performance in core Computer Science courses — mathematics, data structures, algorithms, machine learning.
- **Competitive achievement:** National or international Olympiad medals in Mathematics or Informatics; ICPC regional participation; top performance in national competitive programming contests.
- **Research orientation:** Evidence of independent learning, self-directed projects, GitHub portfolios, or prior exposure to research environments.
- **Long-term commitment:** Demonstrated ability to sustain effort over extended timescales — as evidenced by academic trajectory, prior project completion, and references.
- **Communication capability:** Sufficient English proficiency for academic reading, writing, and international collaboration.

4.2 The Five-Stage Selection Process

Selection for the Research Track is conducted over approximately eight weeks and comprises five sequential stages. Each stage eliminates candidates who do not meet the standard required for the next. The final stage — the Mentor Match — involves the prospective mentor directly, ensuring that the selected resident is genuinely suited to the research direction and working style of their assigned supervisor.

STAGE	NAME	WHAT WE ASSESS
I	CV Screening	Academic record, GPA in core modules, extracurricular achievements, competitive awards, and research portfolio or GitHub activity.

II	Entrance Test	Foundational knowledge in mathematics (probability, linear algebra, calculus), machine learning concepts, and algorithmic thinking. Conducted in multiple parallel sessions.
III	Coding Contest	Algorithmic problem-solving ability — greedy algorithms, dynamic programming, graph traversal, data structures — under timed conditions on the PTIT competitive programming platform.
IV	Paper Presentation	Each candidate is assigned a research paper or advanced technical topic. They must read it independently, prepare a slide presentation, deliver a 15-minute talk, and defend their understanding in a 15-minute Q&A. This stage assesses reading comprehension, technical depth, critical thinking, and communication.
V	Mentor Match	Shortlisted candidates are introduced to prospective mentors. The mentor reviews the candidate's CV and previous-stage performance, conducts a brief online interview, and makes the final selection. Mentors have full discretion at this stage.

5. Programme Structure and Expectations

5.1 Research Track: The Two-Year Arc

The two-year timeline below is intended as a guide; actual pacing will vary by project, resident, and mentor. What remains constant is the expectation of meaningful output at each review point.

Phase I Onboarding	The resident undertakes a structured literature survey in their assigned topic area, develops their reading and note-taking practice, and completes a baseline reproduction of one significant paper. Initial alignment meetings between resident, Lab supervisor, and external mentor establish the working relationship and set expectations.
Phase II Formulation	The resident, guided by their mentor, identifies a research question of sufficient depth and originality. Preliminary experiments are conducted and discussed. The resident delivers an internal presentation to Lab faculty and peers, receiving critical feedback before proceeding.
Phase III First Review	At the end of the first year (approximately), a formal review is conducted. The resident must demonstrate clear research progress — in the form of a workshop submission, preprint, or a well-scoped proposal with preliminary results. Continuation in the programme is contingent on a positive review outcome.
Phase IV Output	The resident conducts full empirical work, writes and revises their paper with mentor feedback, and targets submission to a competitive international venue. Simultaneously, residents on the academic pathway begin preparing their graduate school application materials, with Lab faculty and mentors providing letters of support and strategic guidance where appropriate.

5.2 Commitment and Accountability

All residents sign a commitment document upon acceptance. The programme's standards are high and deliberately maintained. Residents who fail to meet their obligations — through missed deadlines, loss of contact, consistently incomplete work, or lack of teamwork — receive a formal warning; continued failure results in exit from the programme.

This rigour is intentional. It signals to the wider community — and to partner organisations — that a NITS Resident designation carries meaning.

5.3 Regular Review Cadence

Monthly	Resident submits a written progress note to their Lab supervisor covering completed tasks, challenges encountered, and planned next steps.
Quarterly	Lab faculty compile a structured progress report — made available to the external mentor — summarising technical progress, commitment levels, and any issues requiring attention.
End of Year One	Formal mid-programme review. Resident presents to a panel of Lab faculty and (where available) the external mentor. Continuation decision made at this point.
End of Programme	Final assessment of output, quality of publication, and overall contribution. Programme evaluation is conducted to improve future cohorts.

6. Partner Roles and Responsibilities

6.1 Research Mentor

Research Mentors are typically academic faculty or senior researchers at universities or research institutes. They bring the intellectual depth, domain expertise, and scholarly network that transforms a talented student into a genuine researcher. Because Lab faculty handle day-to-day supervision, the Research Mentor's engagement is focused and high-impact rather than high-frequency.

WHAT THE MENTOR PROVIDES

- › Engages on a cadence that fits the project and the resident's development stage
- › Sets the research direction, frames the key questions, and recommends foundational readings
- › Provides the substantive guidance that shapes a 2nd- or 3rd-year student into a capable researcher — this is precisely the stage when good mentorship has the greatest effect
- › Reviews experimental design, analyses results, and gives critical feedback on writing as work matures
- › Optionally hosts residents for short visits, research exchanges, or summer school participation

WHAT THE LAB PROVIDES

- › Handles all recruitment, screening, and vetting of candidates
- › Provides local supervision and day-to-day technical support between mentor meetings
- › Maintains accountability structures: weekly check-ins, progress tracking, warning and exit procedures
- › Manages all scheduling, timezone coordination, and meeting logistics
- › Compiles and shares quarterly progress reports with the mentor
- › Liaises with PTIT for institutional support (travel, exchange programmes, etc.)

We ask mentors to bring their expertise, their questions, and their standards. We ask them to meet the resident where they are — at the beginning of their research life — and invest accordingly. The Lab takes care of the rest.

6.2 Industry Partner

Industry Partners typically engage with the Engineer Track, providing real project environments, technical mentorship, and pathways to internship or full-time employment for outstanding residents. We also welcome industry partners who wish to co-supervise research residents on applied topics of mutual interest.

Industry Partners are invited to:

- Provide project briefs or research challenges for Engineer Track residents.
- Assign a technical supervisor who meets with the resident periodically (monthly or as project needs dictate).
- Offer internship or graduate program pathways for residents who meet the partner's hiring standards.
- Participate in career events, workshops, or guest lectures organized by NITS Lab and PTIT.
- Optionally co-brand programme activities and access the NITS Lab talent pool for early-stage recruiting.

6.3 Research or Academic Institution Partner

Universities and research institutes can partner with NITS Lab at multiple levels — from providing individual Research Mentors to establishing a broader institutional collaboration that includes joint research projects, student exchanges, and co-organized programmes such as summer schools or workshops.

Institutional partnerships may include:

- Formal MOU or Letter of Intent between the partner institution and PTIT.
- Joint supervision arrangements for graduate-bound residents.
- Co-organization of international summer schools or short courses at PTIT's Hadong campus.
- Joint grant applications in areas of shared research interest.
- Facilitated exchanges — short visits by NITS residents to the partner's lab or campus.

7. Existing Partnerships and Broader Context

7.1 Current Research Collaborations

NITS Lab is already engaged in active research collaborations with several international institutions. These relationships provide an existing network into which new partners are welcomed, and demonstrate the Lab's capacity to execute sustained international research partnerships.

A2i2 – Deakin University (Australia)	Ongoing research collaboration including a regular joint Reading Group. A2i2 (Applied AI Institute) is one of Australia's leading applied AI research centres.
University College Cork (Ireland)	Research collaboration in AI and computer science, with joint project development underway.
Adobe Research (United States)	Research collaboration with senior members of Adobe's US research team, contributing to the Lab's work in vision and multimodal AI.
Samsung (Republic of Korea)	Institutional sponsorship supporting the Samsung Training Room and Samsung Research Lab within PTIT's Faculty of IT I, in connection with programmes such as SIC and SSAFY.
Singapore Institute of Technology	Co-organization of the International Summer School on AI for Cybersecurity (July 2026).
Brunel University London (United Kingdom)	Co-organization of the International Summer School on Artificial Intelligence (August 2026).

7.2 Industry Relationships

The Lab maintains active relationships with a range of Vietnamese and international technology companies:

Microsoft Vietnam	Guest lectures, career workshops, and technical certifications (Azure AI, Microsoft AI-900, AI-102).
FPT Software / FPT AI	Internship pathways, joint projects, and alumni network connections.
Viettel	Internship and joint project opportunities for Engineer Track residents.
NVIDIA	Deep Learning Institute (DLI) certifications including Fundamentals of Deep Learning, Computer Vision, and Transformer NLP workshops.
Amazon Web Services	Cloud Practitioner and Machine Learning Specialty certifications; career pathways.
Khantix & NobiSoft	Bootcamp co-organization and Fullstack/AI Dev residency programmes (ongoing).

8. Contact and Next Steps

If you are interested in participating as a Research Mentor, Industry Partner, or Institutional Partner, we would welcome a conversation. There is no obligation at this stage — we are happy to answer questions, share more detail about specific programme elements, or schedule a brief meeting with the programme team.

We would be honoured to have you join the NITS Research Residency Programme.

8.1 Programme Director

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8.2 Suggested Next Steps

Explore	Review this document and the programme flyer. Identify whether the Research Mentor, Industry Partner, or Institutional Partner role is the best fit.
Connect	Send an expression of interest email to long.dh@ptit.edu.vn with a brief description of your background and intended mode of engagement.
Discuss	We will schedule a 30-minute introductory call to answer questions, explore the match, and outline next steps.
Formalise	For Research Mentors: receive and review the shortlisted resident CVs and arrange a Mentor Match interview. For Institutional Partners: discuss MOU or collaboration terms with our faculty team.
Launch	Selected residents begin the programme in the period July–September 2026, with onboarding and initial planning sessions coordinated by Lab staff.

This document is intended for prospective partners of the NITS Research Residency Program. Its contents are confidential and intended solely for the recipient. For questions or to request additional information, please contact Dr. Long Hoang Dang at long.dh@ptit.edu.vn.

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